TECHNOLOGY

AWS Solution Architect

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Container Service

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Learning Objectives

By the end of this lesson, you will be able to:

- Execute a Docker container on an EC2 instance to improve resource utilization
- Push and pull container images from Amazon ECR to implement container-based applications
- Create and manage container clusters using Amazon ECS for various AWS service integration

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Elastic Container Service

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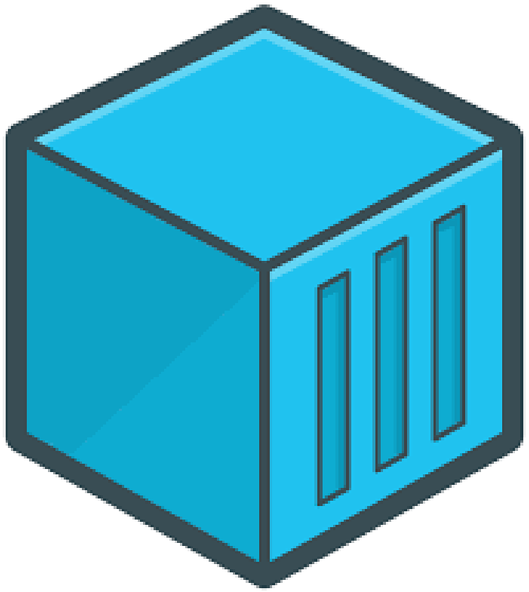
What Is a Container?

A container is a standardized software development unit encompassing all elements necessary for a software application. These elements include code, runtime, system tools, system libraries, and more.



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What Is Docker?

Docker is a tool designed to automate the deployment of applications as lightweight containers, enabling efficient operation across diverse environments.

A Docker container is a streamlined software package that encompasses all necessary dependencies, such as code, frameworks, and libraries, to execute an application effectively.

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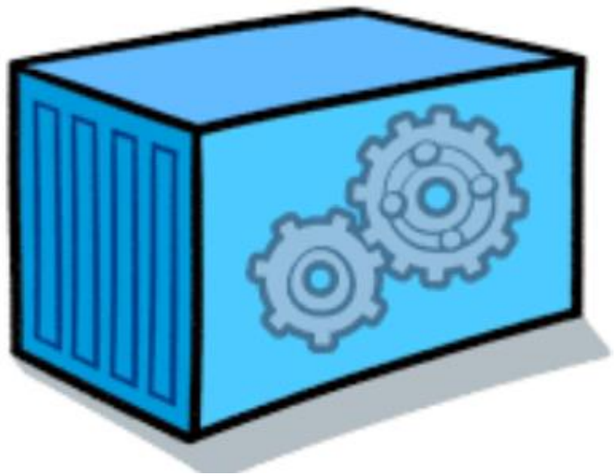
Advantages of a Docker Container



01 They are highly scalable and efficient.

02 They boot up quickly.

03 Docker containers demonstrate high portability.

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Running a Docker Container on an AWS EC2 Instance

Duration:15 min

Problem Statement:

You have been asked to demonstrate the process of running a Hello World Docker container on an AWS EC2 instance for efficient container deployment.

Outcome:

By performing this demo, you will learn how to run the Hello World container on an EC2 instance. This will enhance your understanding of containerization and provide hands-on experience with deploying Docker containers on AWS.

Note: Refer to the demo document for detailed steps:
01_Running_a_Docker_Container_on_an_AWS_EC2_Instance

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Assisted Practice: Guidelines

Steps to be followed are:

1. Set up an EC2 instance
2. Install the Docker on Ubuntu
3. Run the Hello World Container

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Creating a Container Registry Using AWS ECR

Duration:15 min

Problem Statement:

You have been asked to create an AWS ECR container registry, configure Docker on your EC2 instance, and push a Docker image to your ECR repository.

Outcome:

By performing this demo, you will learn how to create a container image repository using AWS ECR and push images into it. This will enhance your skills in managing container images and streamline the deployment process in a cloud environment.

Note: Refer to the demo document for detailed steps:
02_Creating_a_Container_Registry_Using_AWS_ECR

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create an ECR repository
2. Launch an EC2 instance
3. Install Docker on the EC2 instance
4. Create and push the Docker image to the repository

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Quick Check

You are working on a project where you need to ensure that your application runs consistently across different environments, from development to production. Which of the following tools would be the most appropriate to achieve this, considering the advantages of containerization?

A. Virtual Machines

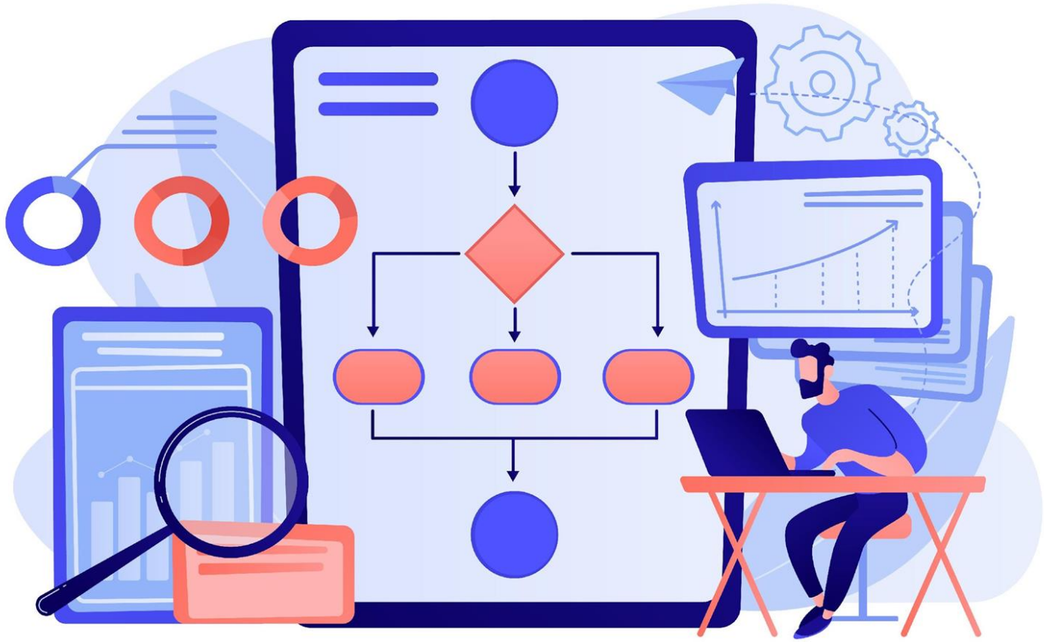
B. Docker

C. Traditional Deployment Scripts

D. Cloud-based Hosting Services

What Is Container Orchestration?

Container orchestration automates scheduling, development, networking, scaling, health monitoring, and management of containers.



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Container Orchestration Tools in AWS

The AWS services used for container orchestration are as follows:

Amazon ECS

Amazon EKS

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Amazon Elastic Container Registry (ECR)

Amazon Elastic Container Registry (ECR) is a fully managed Docker container registry that simplifies developers' tasks of storing, managing, and deploying Docker container images.

AWS ECR

It is integrated with Amazon Elastic Container Service (ECS) to streamline the container deployment process.

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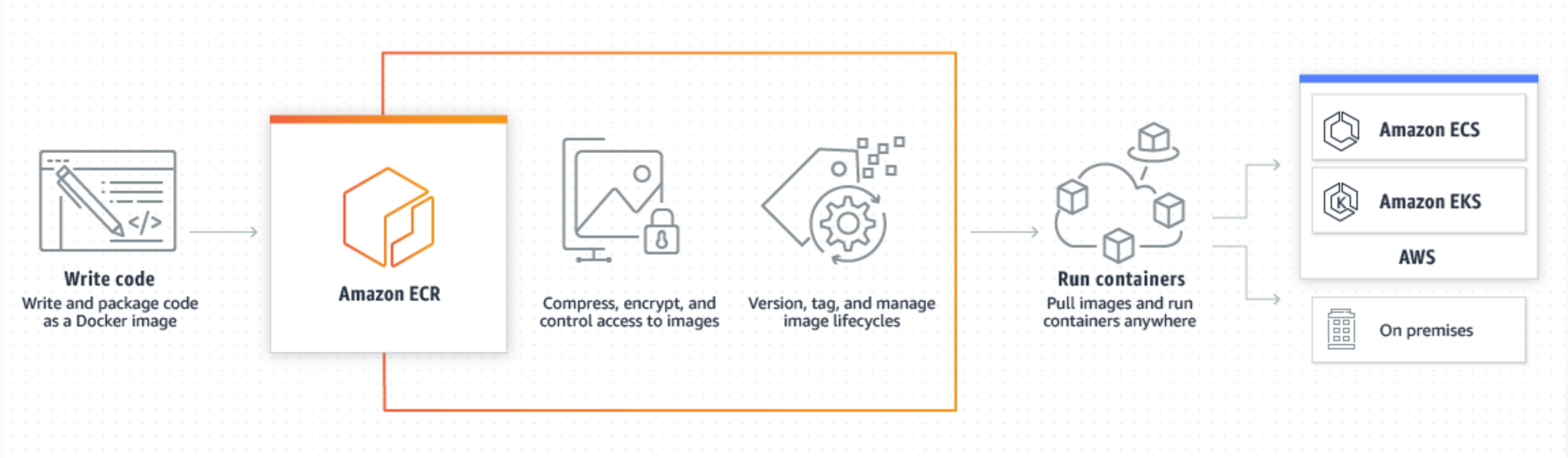
Amazon ECR Workflow

This image illustrates the role of Amazon ECR in the container deployment process:



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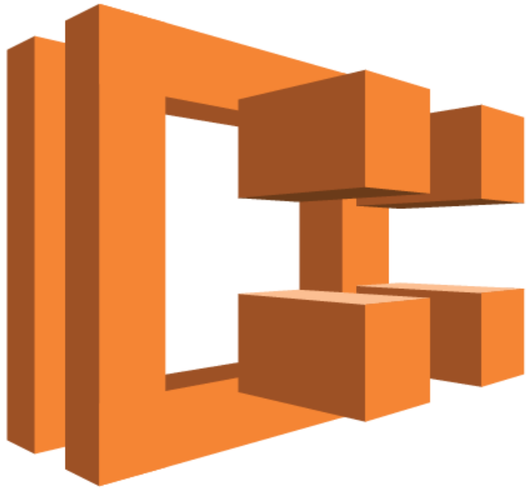
Amazon Elastic Container Service

Amazon Elastic Container Service (Amazon ECS) is a highly scalable and rapid container management service that deploys and manages containers on a cluster.



ECS is integrated with the broader AWS platform, offering a secure and user-friendly solution for executing container workloads in the cloud.

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ECS: Features

- Amazon ECS fully supports Docker, enabling users to run and manage Docker containers effectively.
- It provides flexibility for users to integrate and extend their service by making API calls.
- It automatically restores failed containers, ensuring consistent application availability and reliability.
- It facilitates cluster scheduling based on diverse requirements, such as configuring clusters according to resource needs like CPU or RAM.

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Quick Check

Your team is tasked with deploying a set of microservices that need to scale quickly based on demand. You're considering using Amazon ECS due to its AWS integration and ability to manage containers efficiently. What key benefit of Amazon ECS would make it particularly suitable for your team's needs?

A. It provides a completely independent environment from AWS.

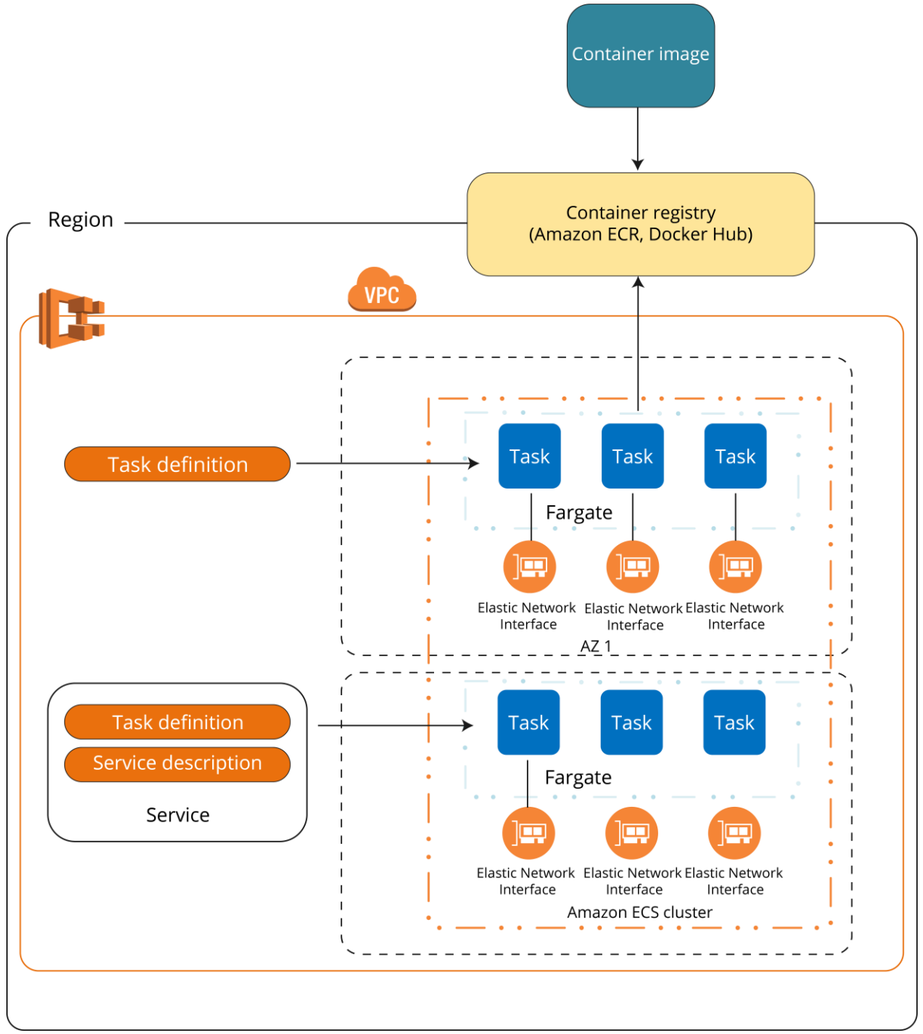
B. It offers a scalable and rapid container management service integrated with AWS.

C. It requires manual scaling of containers, giving you full control over resource allocation.

D. It is only useful for small-scale applications with minimal traffic.

ECS Architecture

The image below illustrates how Amazon ECS employs AWS Fargate to execute containers:



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ECS: Components

01

Cluster:
It is a logical grouping of tasks or services within ECS.

02

Container:
It is a standardized software unit containing all the components necessary for an application to function.

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ECS: Components

03

Task Definition:
It is a JSON-formatted text file describing one or more containers that compose an application.

04

Task:
It is the instantiation of a task definition within an ECS cluster.

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ECS: Components

05

Service:
The purpose of a service is to concurrently execute and uphold the desired count of tasks within an Amazon ECS cluster.

06

Container Agent:
It operates on each container instance within an Amazon ECS cluster and transmits details about ongoing tasks and the container's resource utilization to Amazon ECS.

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Elastic Container Service (ECS)

Amazon ECS can be launched in the following modes:

Fargate launch

EC2 launch

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When to Use Fargate Launch Type?

The Fargate launch type is well-suited for the following workloads:

01 Large workloads that require low operational overhead

02 Small workloads that have an occasional burst

03 Batch workload that require high operational overhead

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When to Use EC2 Launch Type?

The EC2 launch type is ideal for the following workloads:

01 Workloads demanding consistently high CPU core and memory usage

02 Large workloads requiring cost optimization

03 Applications necessitating access to persistent storage

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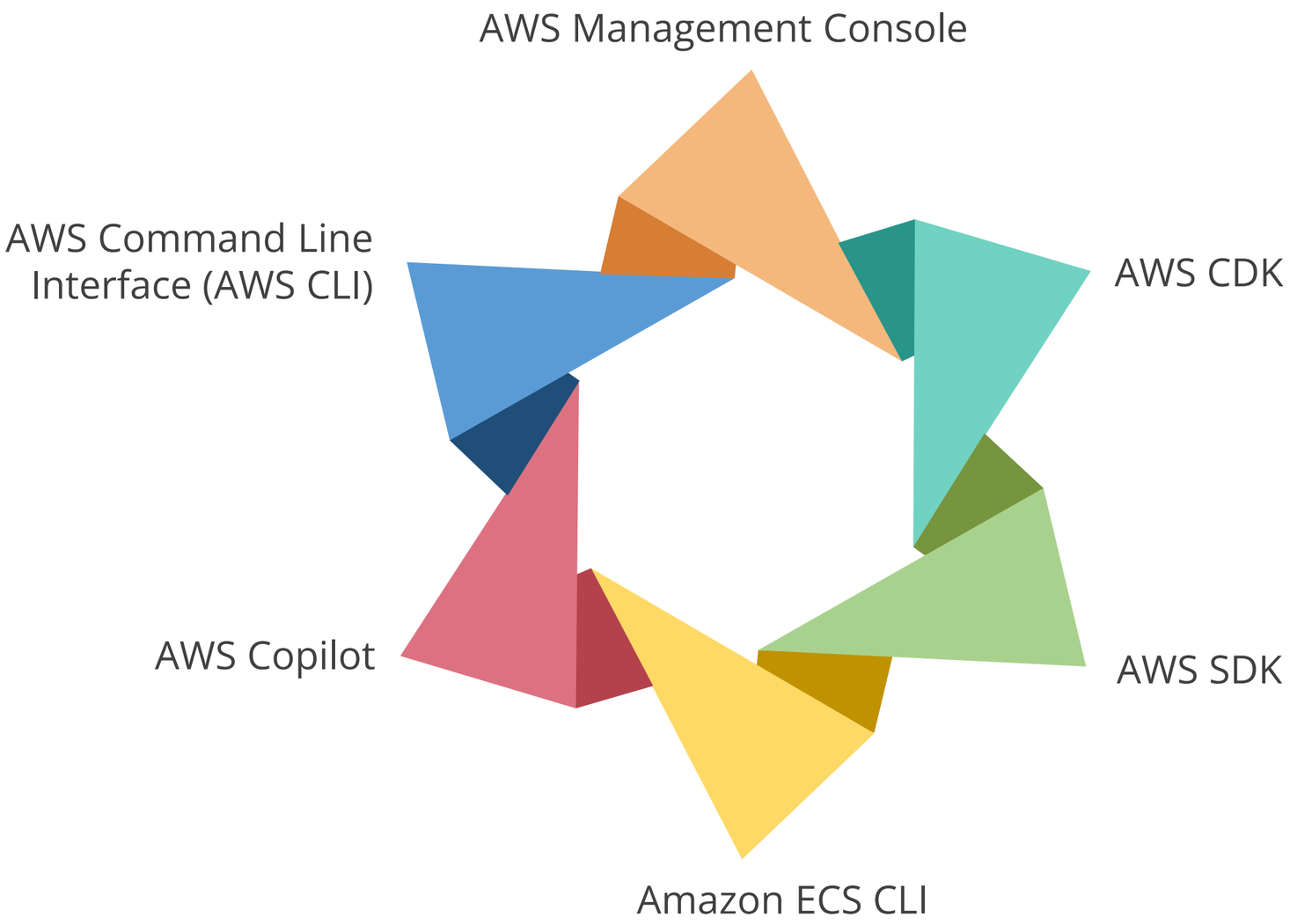
How to Access ECS?

ECS can be accessed from the following interfaces:



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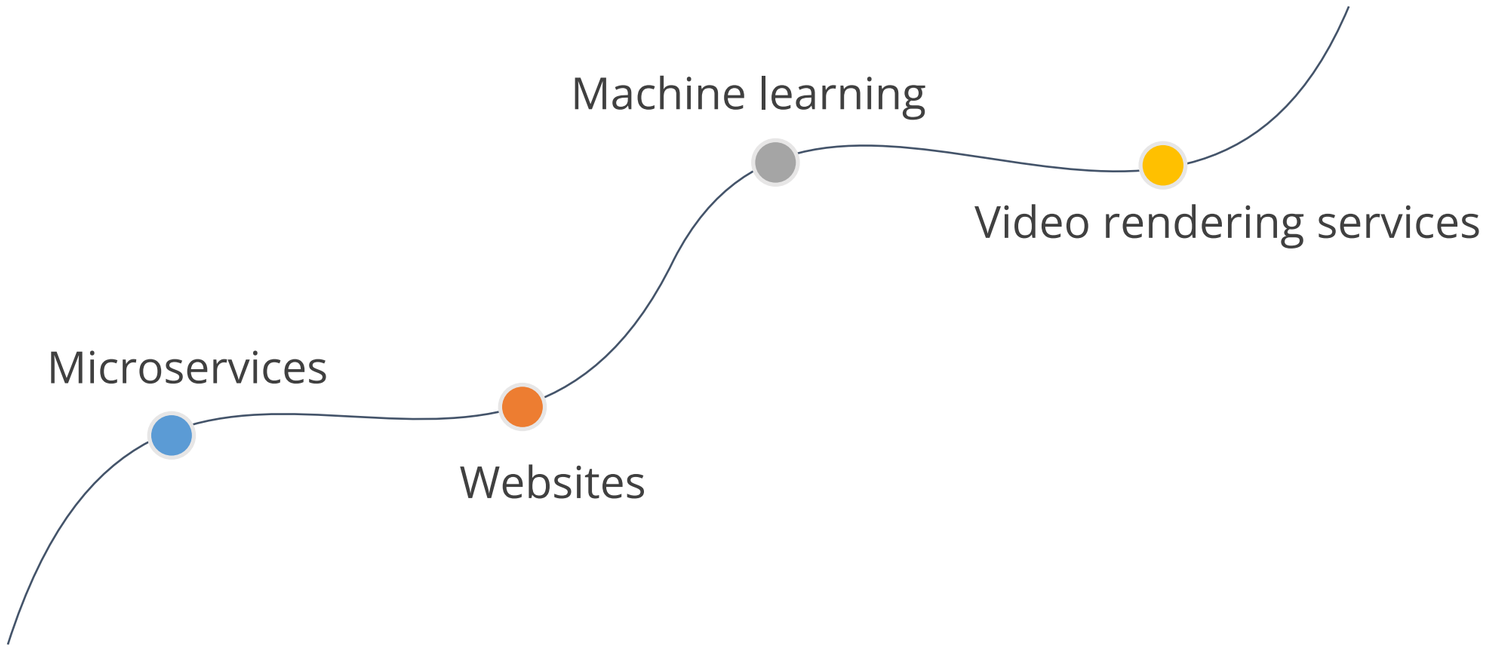
ECS: Use Cases

Amazon ECS is suited for a variety of use cases, including:



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Companies Using Amazon ECS

The following companies use Amazon ECS:

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Creating an EC2 Cluster with Auto Scaling

Duration:15 min

Problem Statement:

You have been asked to demonstrate how to create an ECS cluster with auto-scaling and Container Insights for enhanced monitoring and automatic adjustment of resource allocation based on demand.

Outcome:

By performing this demo, you will learn how to create an ECS cluster with auto scaling and Container Insights, and how to run a task definition for a Nginx container on the cluster. This will enhance your ability to manage scalable containerized applications and monitor their performance on AWS.

Note: Refer to the demo document for detailed steps:
03_Creating_an_ECS_Cluster_with_Auto_Scaling

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Assisted Practice: Guidelines

Steps to be followed are:

1. Create an ECS cluster
2. Create a task definition
3. Run the task definition on the cluster

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Quick Check

You are deploying a new service on Amazon ECS and need to specify the configuration for your containers, including resources and environment variables, in a structured format. Which of the following describes the JSON file where you define this configuration?

A. Cluster

B. Task

C. Container

D. Task Definition

Key Takeaways

- Containers are software packages that are used to execute an application in any environment.
- Docker is a containerization platform that enables users to package applications into containers.
- The deployment, management, scaling, and networking of containers are all automated through container orchestration.
- Amazon ECS is a container orchestration service that supports Docker containers and allows users to easily run and scale containerized applications on AWS.

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Monitoring Insights for ECS Cluster Using CloudWatch

Duration: 30 mins

Project agenda: To monitor an Amazon Elastic Container Service (ECS) cluster using CloudWatch Container Insights

Description: Your company is going to deploy a container application using ECS clusters. As a cloud architect, you should be able to monitor the cluster's health by measuring various metrics such as CPU utilization, memory utilization, task count, and so on.

Perform the following:

1. Monitor the cluster using the CloudWatch container

LESSON-END PROJECT

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Thank You

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